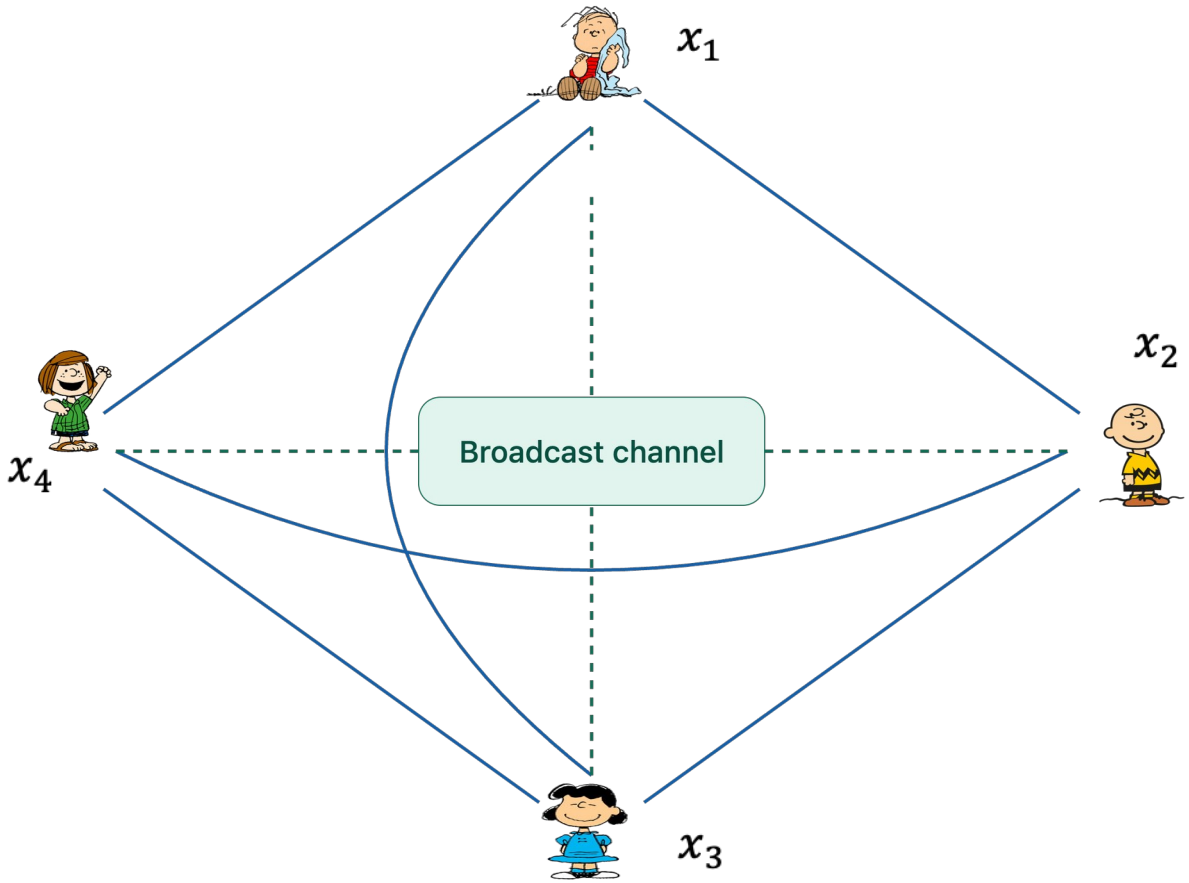


Statistical MPC with a Constant Communication Overhead

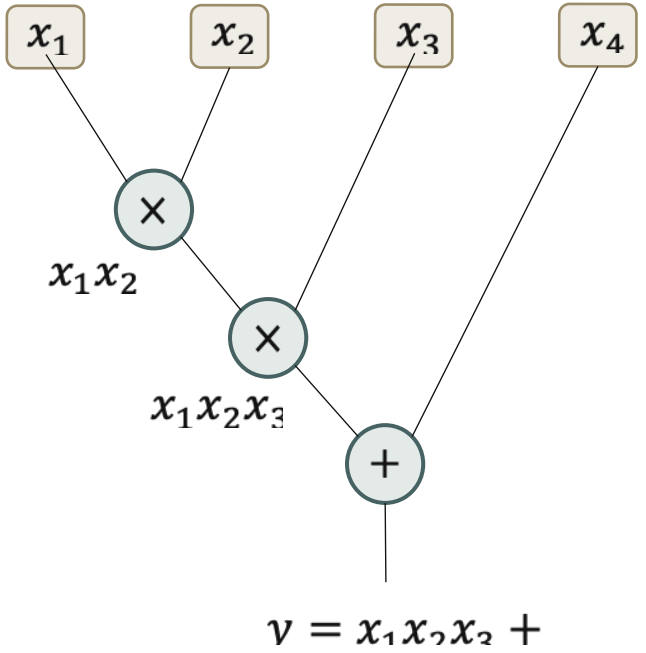
Ashish Choudhury, Ivan Damgård, Shravani Patil, Arpita Patra · TCC 2026

Statistical MPC with GOD is possible whenever t < n/2 [RB89]



$f: \mathbb{F}^n \Longrightarrow \mathbb{F}$

Arithmetic ckt abstraction





T. Rabin



M. Ben Or

Any f can be computed with **statistical security** in the presence of $t < \frac{n}{2}$ **malicious** corruptions

Verifiable Secret Sharing and Multiparty Protocols with Honest Majority (Extended Abstract). STOC 1989: 73-85

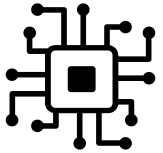
Setting: t < n/3 adaptive, malicious corruptions; statistical security



Type of corruption

Threshold
t < n/3, i.e. n = 3t + 1

Non-threshold



Computational power of the adversary

Unbounded
Statistical security

Bounded



Type of adversarial behavior

Semi-honest

Malicious
Adaptive corruptions



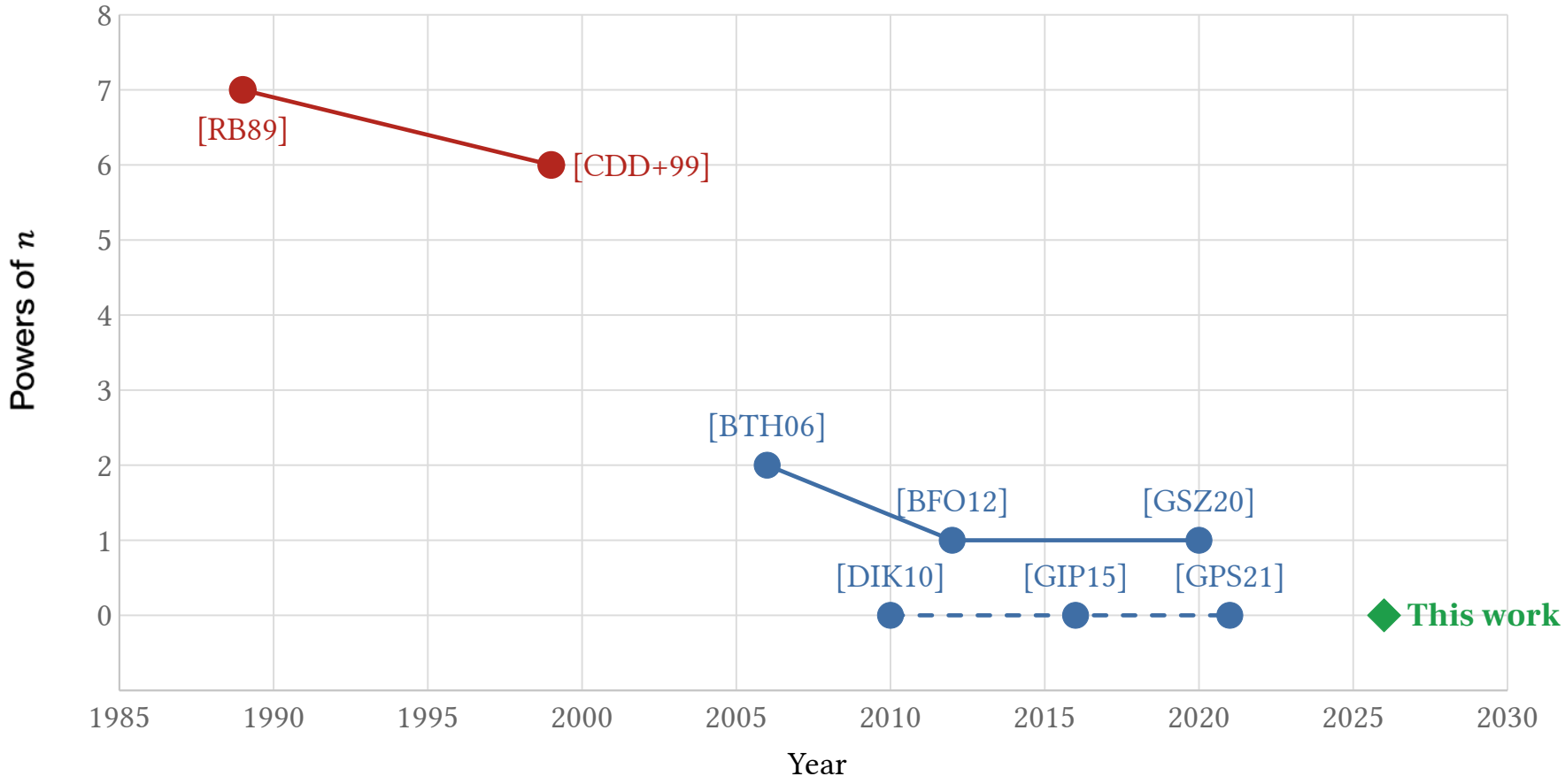
Network type

Synchronous
Private channels + broadcast

Asynchronous

Goal: guaranteed output delivery (GOD)

Per-gate cost fell from n^7 to O(1), but only by giving up O(D) rounds



Round-efficient
O(D) rounds

Communication-efficient
D + poly(n) rounds
solid: n = 2t + 1
dashed: n = 2t + Θ(n)

This work
O(1) per gate, O(D) rounds

D: multiplicative depth of the circuit

All points: statistical security with GOD ([DIK10], [GIP15], [GPS21] via the [IKPSY16] compiler). [GPS21] reaches O(1) per gate in O(D) rounds only with abort.
[BFO12]: Eli B, Serge F, Rafail O.: Near-Linear Unconditionally-Secure Multiparty Computation with a Dishonest Minority. CRYPTO 2012
[GSZ20]: Guaranteed Output Delivery Comes Free in Honest-Majority MPC. Vipul Goyal, Yifan Song, Chenzhi Zhu. CRYPTO 2020
[RB89] Rabin & Ben-Or, STOC 1989 · [CDD+99] Cramer et al., EUROCRYPT 1999 · [BTH06] Beerliová-Trubíniová & Hirt, TCC 2006 · [DIK10] Damgård, Ishai & Krøigaard, EUROCRYPT 2010 · [GIP15] Genkin, Ishai & Polychroniadou, CRYPTO 2015 · [IKPSY16] Ishai et al., CRYPTO 2016 · [GPS21] Goyal, Polychroniadou & Song, CRYPTO 2021

O(1) per gate is possible only with $s = \Theta(n)$ extra parties [DLN19]

LOWER BOUND [DLN19]

$\Omega(n \cdot |C| / s)$

Total communication, $n = 2t + s$ parties, statistical security

Even for passive corruptions; circuits of size linear in the input

$s = \Theta(1)$ $\longrightarrow$ $\Omega(n)$ per gate

Matched with GOD by [BFO12], [GSZ20]

$s = \Theta(n)$ $\longrightarrow$ $\Omega(1)$ per gate

Open: constant overhead with GOD and O(D) rounds

Question: can statistical MPC with GOD and O(D) rounds reach O(1) overhead when $n = 2t + s$ with $s = \Theta(n)$?

This work: yes, for $s = t + 1$, i.e. $n = 3t + 1$

Overhead = communication per gate.

[BFO12] Ben-Sasson, Fehr & Ostrovsky, CRYPTO 2012 · [DLN19] Damgård, Larsen & Nielsen, CRYPTO 2019 · [GSZ20] Goyal, Song & Zhu, CRYPTO 2020

Every earlier O(1) protocol gives up either O(D) rounds or GOD

Protocol	Parties	Per gate	Rounds	Security	Circuit
[DIK10], [GIP15]✦[IKPSY16]	$n = 2t + \Theta(n)$	O(1)	**D ✦ poly(n)**	GOD	SIMD
[GPS21]	$n = 2t + \Theta(n)$	O(1)	O(D)	**Abort**	Single
[GPS21]✦[IKPSY16]	$n = 2t + \Theta(n)$	O(1)	**D ✦ poly(n)**	GOD	Single
This work	$n = 3t + 1$	**O(1)**	**O(D)**	**GOD**	SIMD

Trade-off: this work tolerates fewer corruptions ($t < n/3$) and needs SIMD circuits; a general circuit compiles to SIMD form at a $\log|C|$ factor [DIK10].

SIMD: many copies of the same circuit evaluated in parallel. GOD: guaranteed output delivery. Red marks the property each protocol gives up.

[DIK10] Damgård, Ishai & Krøigaard, EUROCRYPT 2010 · [GIP15] Genkin, Ishai & Polychroniadou, CRYPTO 2015 · [IKPSY16] Ishai, Kushilevitz, Prabhakaran, Sahai & Yu, CRYPTO 2016 · [GPS21] Goyal, Polychroniadou & Song, CRYPTO 2021

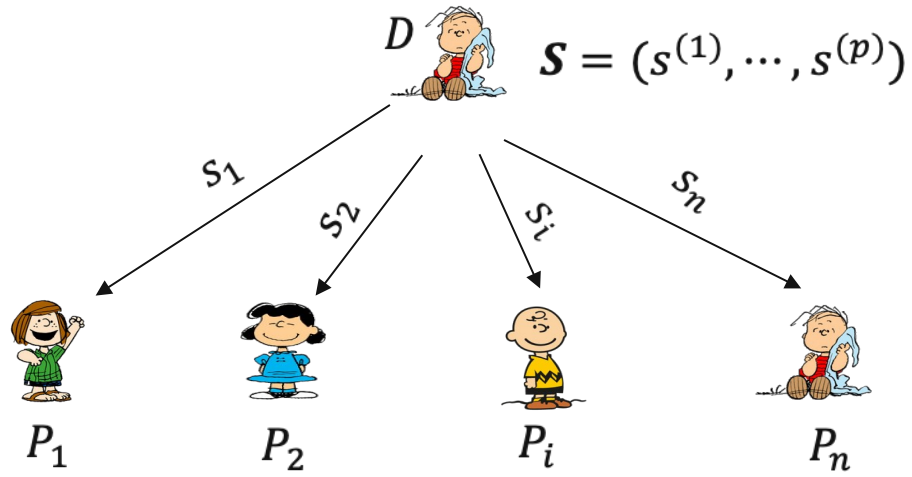
O(1)	O(D)	GOD
field elements per gate	expected rounds	guaranteed output delivery

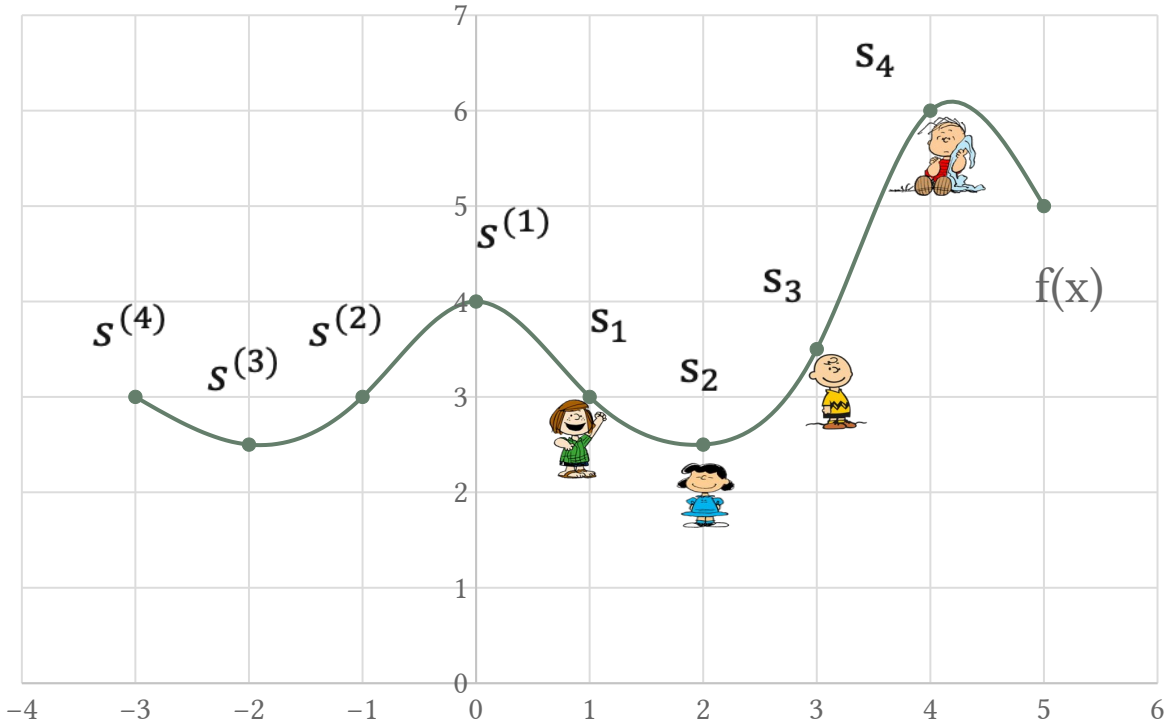
$n = 3t + 1$ parties, statistical security, malicious adversary

$\Theta(n^7)$ SIMD copies; total $O(c_M n^7 + n^8)$ field elements (Theorem 7.1)

Our result: O(1) per gate, O(D) rounds and GOD, at $n = 3t + 1$

Packed sharing [FY92] puts p secrets in one polynomial, at O(1) each





- Less than $t + 1$ shares: no info' about S
- $t + p$ or more shares: complete S

Communication overhead
$\mathcal{O}(1)$, if $p = \Theta(n)$

$d \stackrel{\text{def}}{=} t + p - 1$

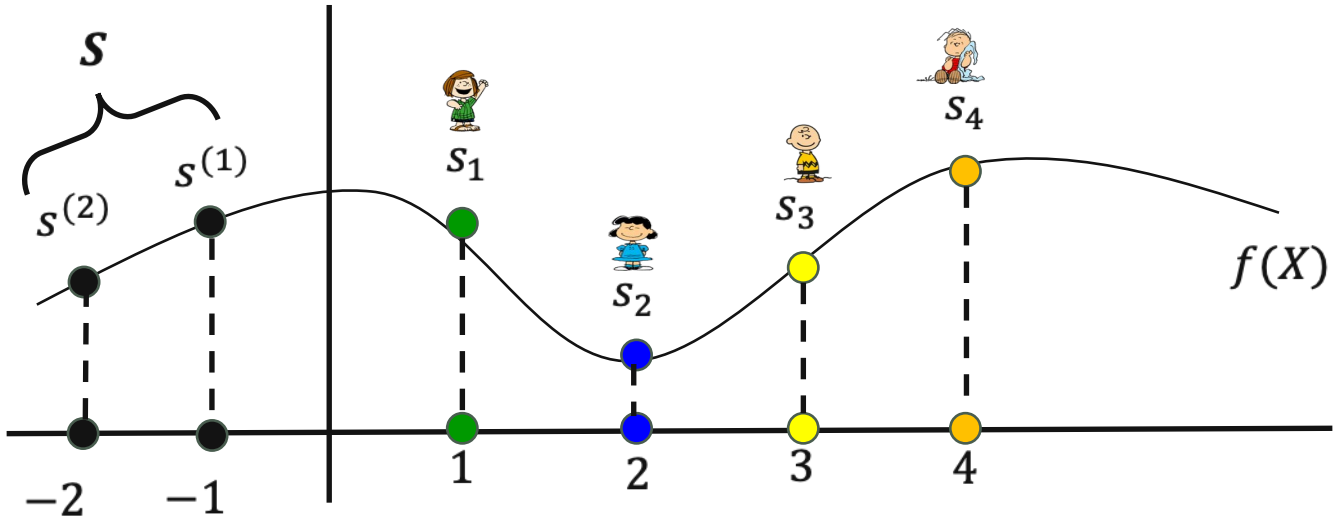
$[S]_d$

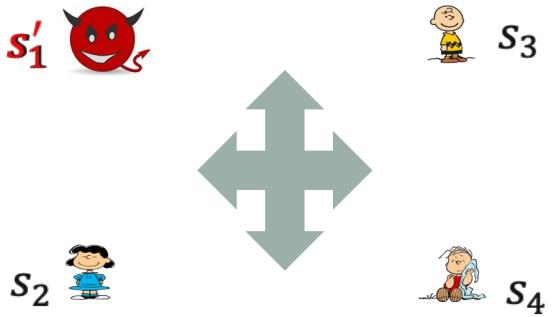
At n = 3t ✦ 1, shares need MACs for robust reconstruction

Number of Parties	Packing Parameter	Degree of Poly
$n = 3t + 1$	$p = \frac{t}{4} + 1$	$d = t + p - 1 = t + \frac{t}{4}$

Authenticated Packed Secret Sharing
(for **robust** reconstruction)

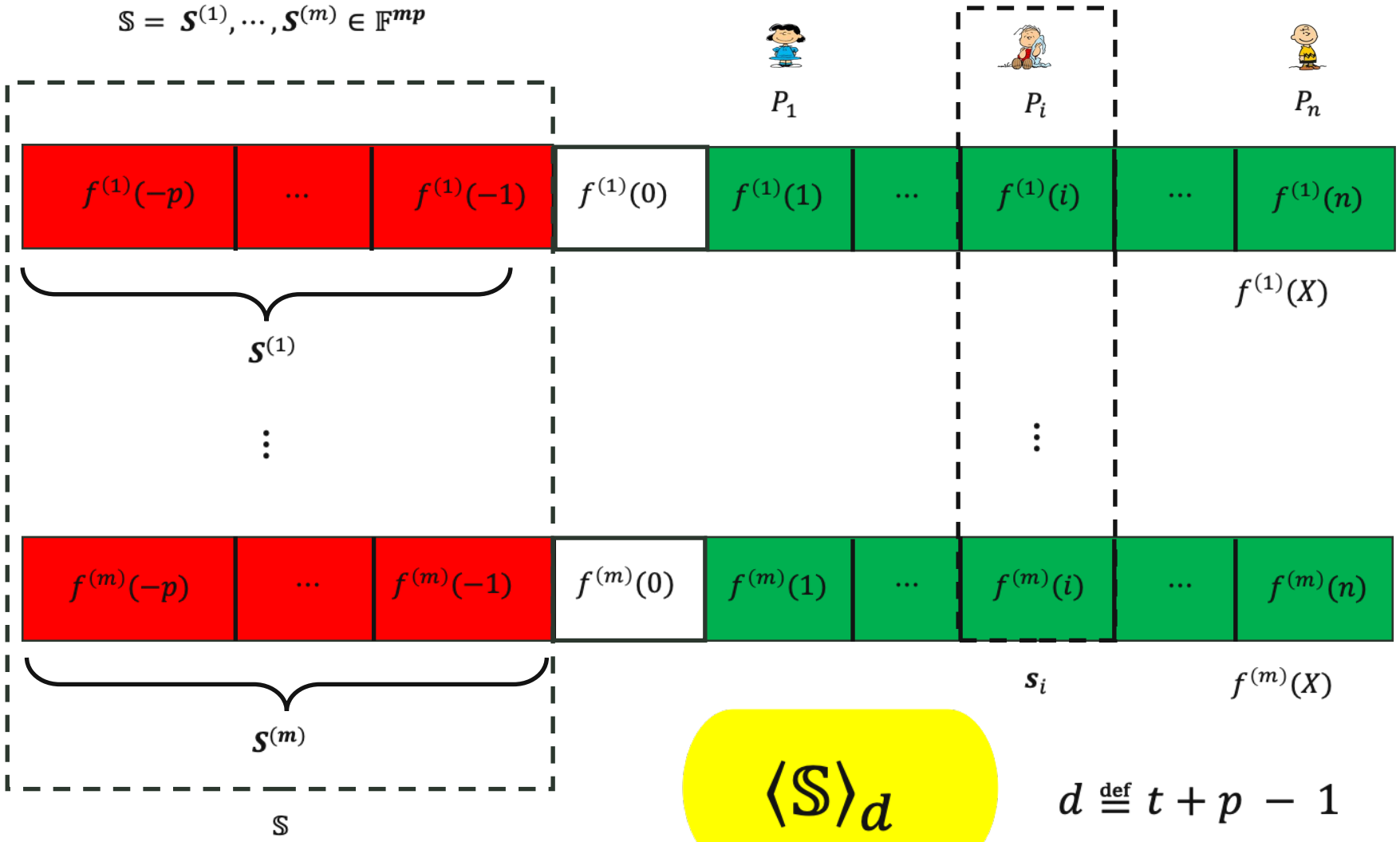
Ex: $t = 1, n = 4, p = 2, d = 2$

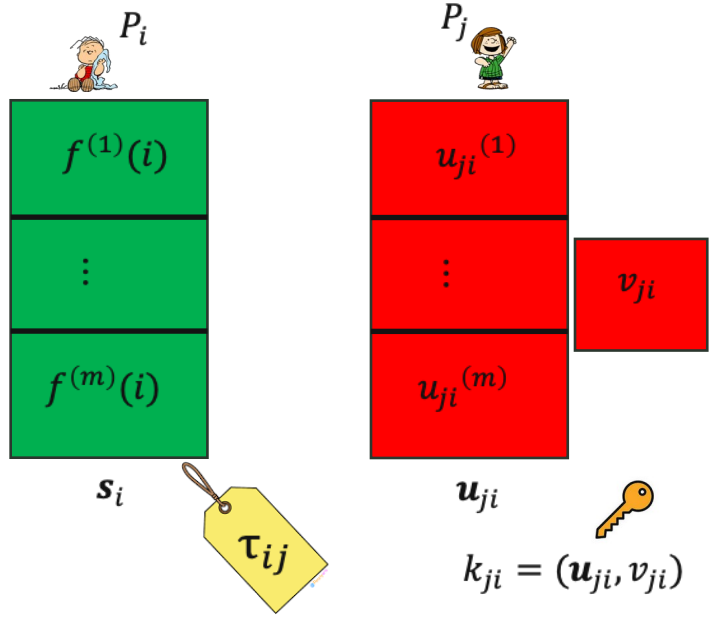




Shares are **authenticated** under pairwise MACs for robust reconstruction

APSS: one MAC tag authenticates each party's share vector





$\tau_{ij} = \boldsymbol{u}_{ji} \cdot \boldsymbol{s}_i + v_{ji}$

$\langle \mathbb{S} \rangle_d$

$d \stackrel{\text{def}}{=} t + p - 1$

Robust reconstruction of $\mathbb{S}$ as each share-vector is authenticated

Two building blocks: a constant-overhead VSS and a robust multiplication

1 VERIFIABLE SECRET SHARING

O(1) overhead per secret



$\Theta(n^5)$ secrets at once, p^2 per bivariate polynomial



One MAC tag authenticates a whole share vector



At most two restarts: O(1) expected rounds

2 MULTIPLICATION OVER APSS

O(1) overhead per product



Beaver multiplication on authenticated packed sharings



Degree t + t/2: **detect, then identify** the cheaters



No restart: sacrificed triples keep opened values private

Circuits: the protocol targets SIMD circuits; a general circuit compiles to SIMD form [DIK10] at a log|C| factor.

[DIK10] Damgård, Ishai & Krøigaard, EUROCRYPT 2010

VSS reaches O(1) overhead per secret in four phases, restarting at most twice

1 SHARING

Find a public Core



Dealer packs p^2 values per bivariate polynomial; parties sign columns (IC signatures)

2 VERIFICATION

Spot-check consistency



Dealer opens a random linear combination; consistency holds w.h.p.

3 NON-CORE REPAIR

Repair outsiders' shares



Core parties send common points; outsiders interpolate

4 TAG GENERATION

Generate MAC tags



Tags from twisted sharings; complaints resolved in public

Restart 1: no Core forms

back to phase 1



Non-confirming parties join D_p, shares fixed to 0; an honest dealer then succeeds

Restart 2: complaints remain

back to phase 4



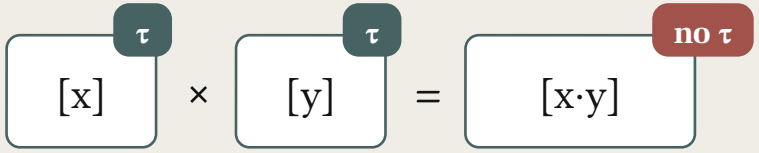
$\geq t - p + 2$ corrupt parties (C_t) identified and zeroed; tags cannot fail again

O(1) per secret needs $\Theta(n^5)$ secrets per dealer, which amortizes the IC-signature and masking costs. Packing parameter $p = t/4 + 1$, degree $d = t + p - 1$.

Builds on: [BFO12] Ben-Sasson, Fehr & Ostrovsky, CRYPTO 2012 and [JLS24] Ji, Li & Song, CRYPTO 2024 (twisted sharings; [JLS24] has a similar VSS structure) · [RB89] Rabin & Ben-Or, STOC 1989 (information checking)

Degree reduction can't correct t errors, so it detects them and finds the cheaters

THE OBSTACLE



MACs are linear, not multiplicative

The product sharing has degree **t ✦ t/2**, and parties cannot compute its MAC tags.

Without tags, **3t ✦ 1** shares cannot correct **t** errors at that degree.

At degree t ✦ t/2, unique decoding falls short



t errors possible

3t/4 correctable



Product sharing: n = 3t + 1 shares, up to t of them wrong (t = 4 shown)

1 Decode
Each P_i corrects up to t/2 errors

≤ t/2 errors

Correct output
Distance t + t/2 + 1 rules out a wrong codeword

> t/2 errors

drop, decode again

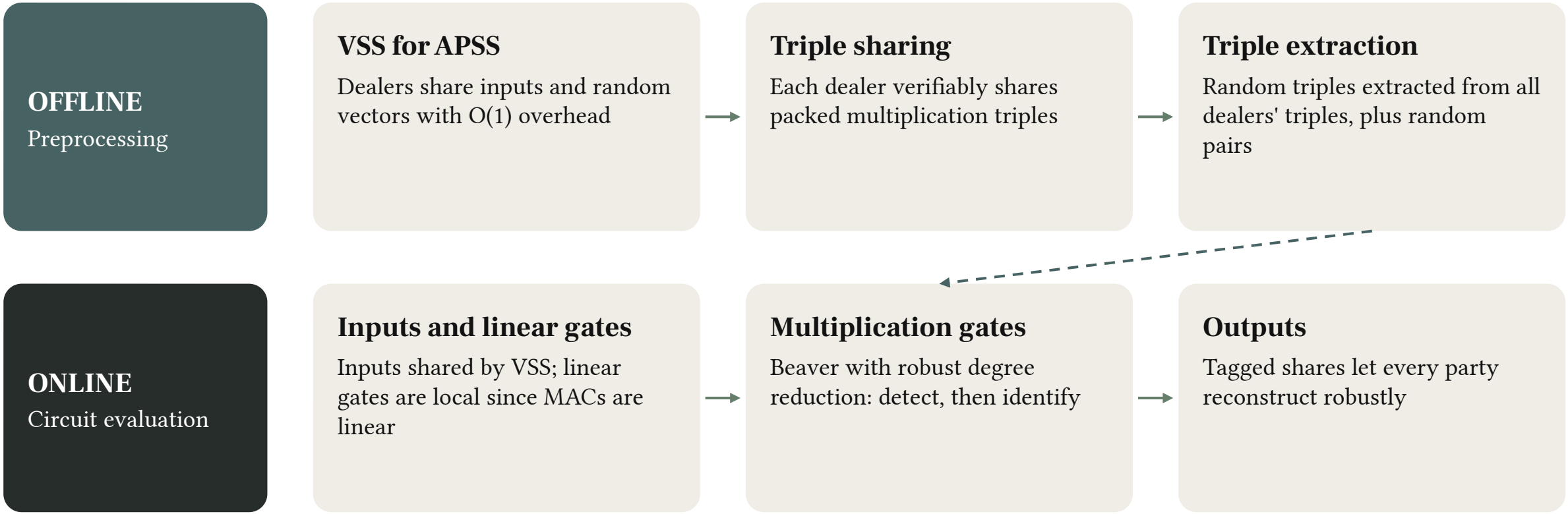
2 Detect
Decoding fails: P_i complains publicly

3 Identify
Open values masked with sacrificed triples; find the > t/2 wrong shares

No restart: one complaint exposes more than t/2 cheaters at once

Builds on: [Bea91] Beaver, CRYPTO 1991 (multiplication triples) · [DPPR25] Damgård, Patil, Patra & Roy, ePrint 2025/1206 (degree reduction for packed sharing)

Offline, the VSS builds packed triples; online, Beaver multiplication uses them



Preprocessing extends the triple-generation framework of [CP15] from polynomial VSS to three-dimensional APSS. Every block is robust and has O(1) overhead, so the whole protocol keeps GOD and O(D) rounds.

[CP15] Choudhury & Patra, ICDCN 2015

Takeaways and open problems

RESULT
$n = 3t + 1$

O(1) per gate

O(D) rounds

GOD guaranteed output

REUSABLE TOOLS

Constant-overhead VSS
for authenticated packed sharings

Robust degree reduction
detect errors, then identify cheaters

OPEN PROBLEMS

1

Fewer secrets?
VSS needs $\Omega(n^5)$ secrets for O(1) overhead

2

Smaller circuits?
O(1) with O(D) rounds and GOD needs $\Theta(n^7)$ SIMD copies

3

Fewer rounds at $n = 2t + \Theta(t)$?
O(1) with GOD costs O(D + poly(n)) rounds; can it be O(D)?